

Deep learning

life cycle of Data science:

1. *read the data*

2. *Preprocess the data*

3. *EDA*

- *Data analysis*
- *Encoding*
- *Scaling*
- *Transformation if required*
- *Feature selection*

4. *Model development*

- *ML*
- *DL*
- *NLP*
- *GenAI*

5. *Tuning*

- *ML hyperparameters*
- *DL hyperparameters*
- *NLP hyper parameters*

6. *Model evaluation*

- *ML evaluation*
- *DL evaluation*
- *NLP evaluation*

7. *Deployment*

1. *ANN : Artificial Neural Network*

- *ALL type of ML models we can achieve using ANN*

2. *CNN : Convolution Neural Network*

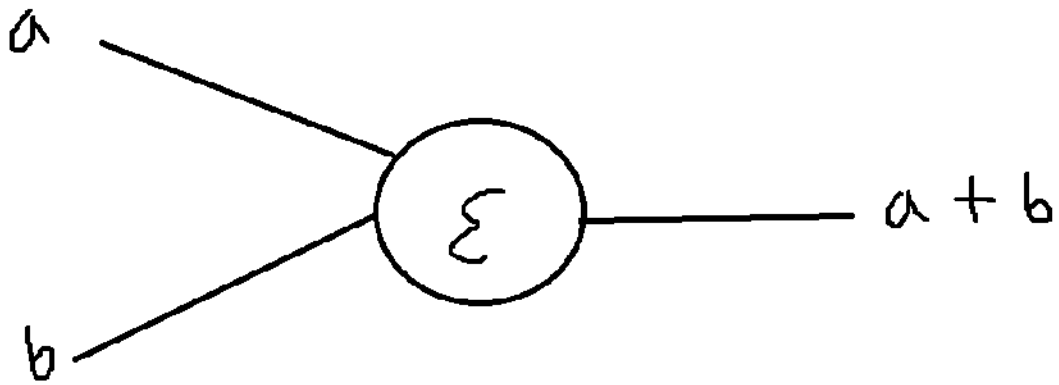
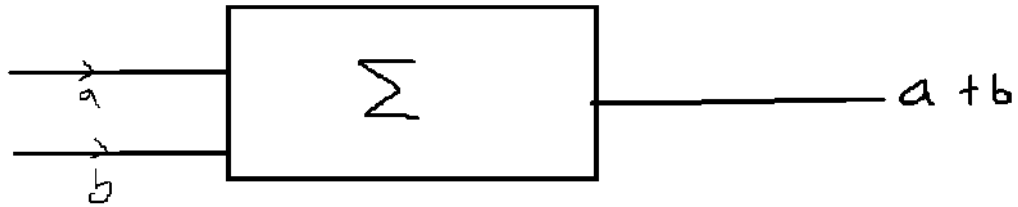
- *Object detection(mainly on Images)*

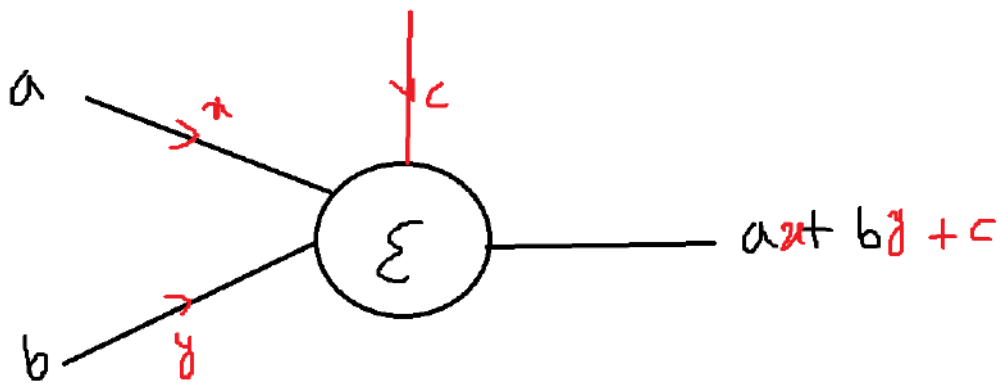
3. *RNN: Recurrent Neural Network*

- *Text related use cases*

1. ANN : Artificial Neural Network

Neuron :





Logistic Regression:

$$y = \frac{1}{1 + e^{-(b_0 + b_1 x_1)}}$$

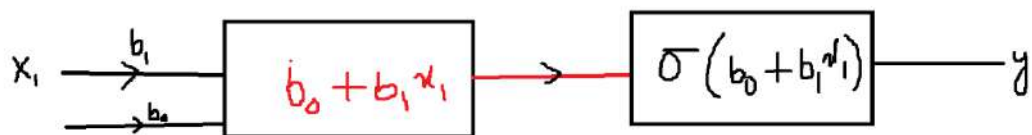
Step – 1: b_0 , b_1 and x_1

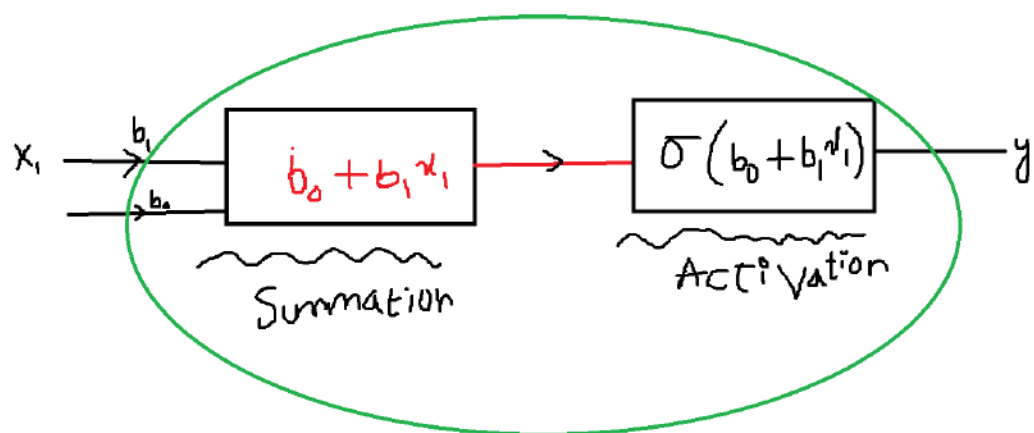
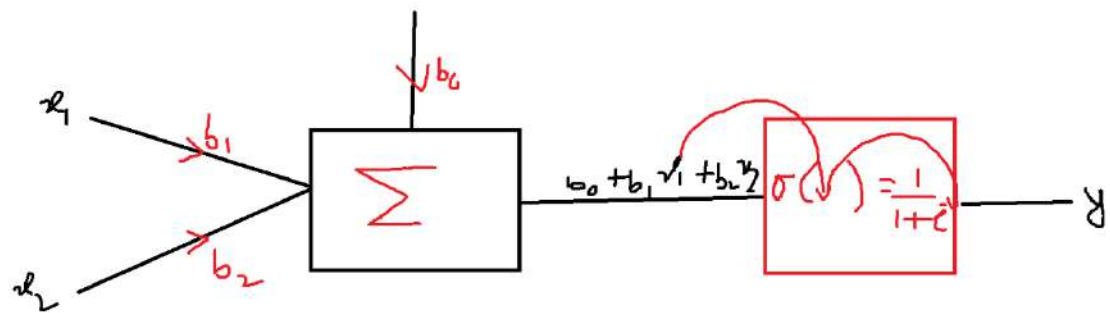
Step – 2: $b_0 + b_1 * x_1$

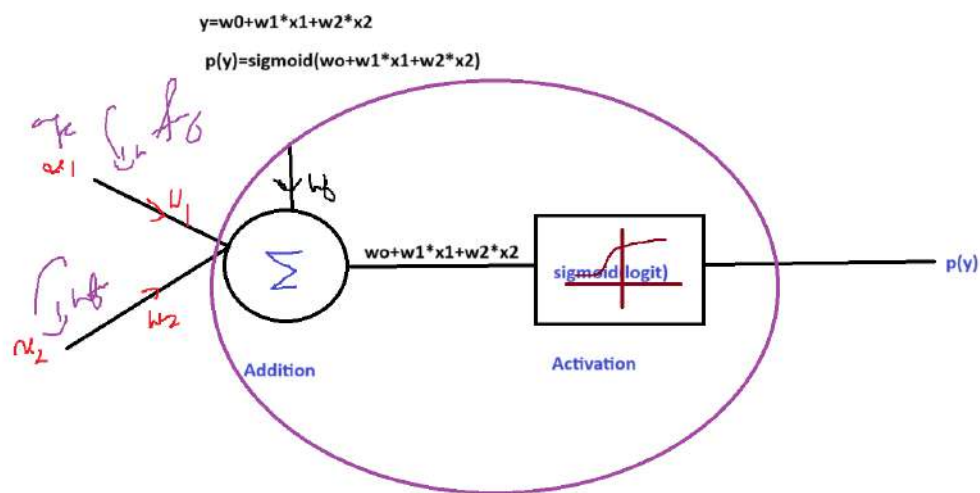
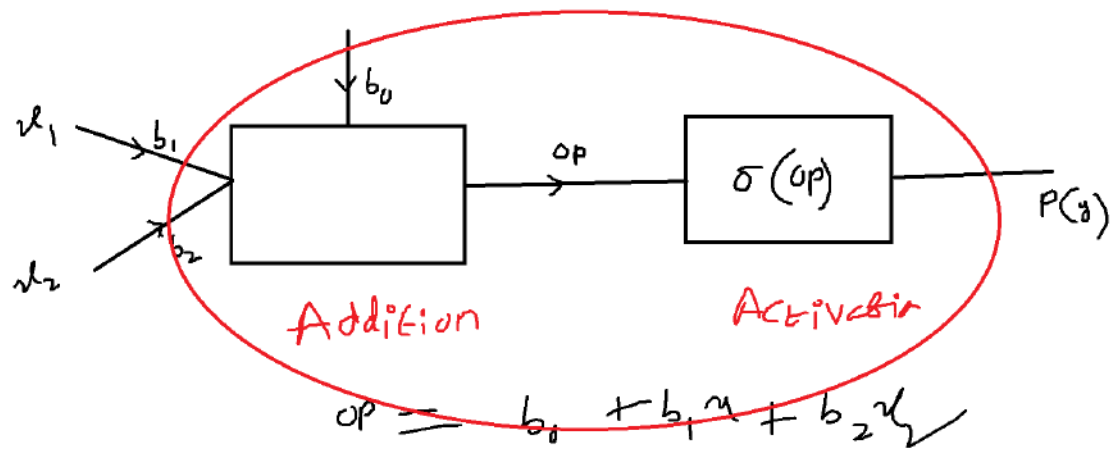
Step – 3: Sigmoid function = $\text{Sigmoid}(b_0 + b_1 * x_1)$

Step – 4: finally we got the y

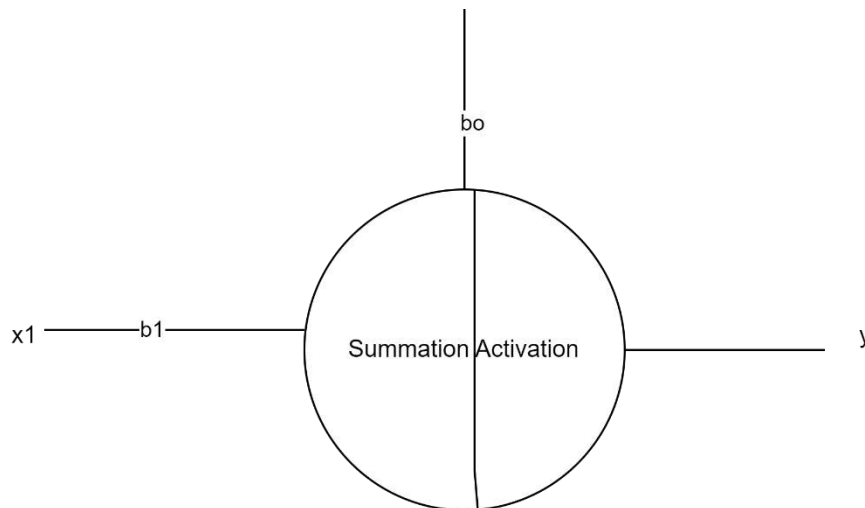
$$y = \frac{1}{1 + e^{-(b_0 + b_1 x_1)}}$$







- *In a Single Neuron both Summation and Activation happens*
- *Sigmoid operation is one of the Activation method*
- *inputs are information gather then understand the pattern these patterns will pass to next neurons*
- *its similar to human analogy, human brain also have billions of nuerons interconnected , and exchange the information*



$$Y = \text{Activation}(bo + b1 * x1)$$

Now we will Create the Network

Three types of layers available

- 1) *Input layer*
- 2) *Hidden layer*
- 3) *Output layer*

Input layer

- *Input layer is used to pass the inputs*
- *Input layer consist neurons*

- The number of neurons equal to number of inputs
- For example in wine quality datasets the number inputs = number of columns
- there are 11 input columns are there so we have 11 neurons
- For example an image size has $28 * 28 = 784$ pixels we have
- so that we required 784 neurons
- Input neurons used only for passing the inputs
- While passing it will not multiply with any weights
- So in the input neurons no summation and no activation

Hidden layers

- Hidden layers are between input and output layer
- It is called hidden because it does not takes the inputs directly
- The input value for hidden layer is: input data multiply with weight upon activation
- hidden layers are used to identify the patterns
- hidden layer neurons has bias and weights

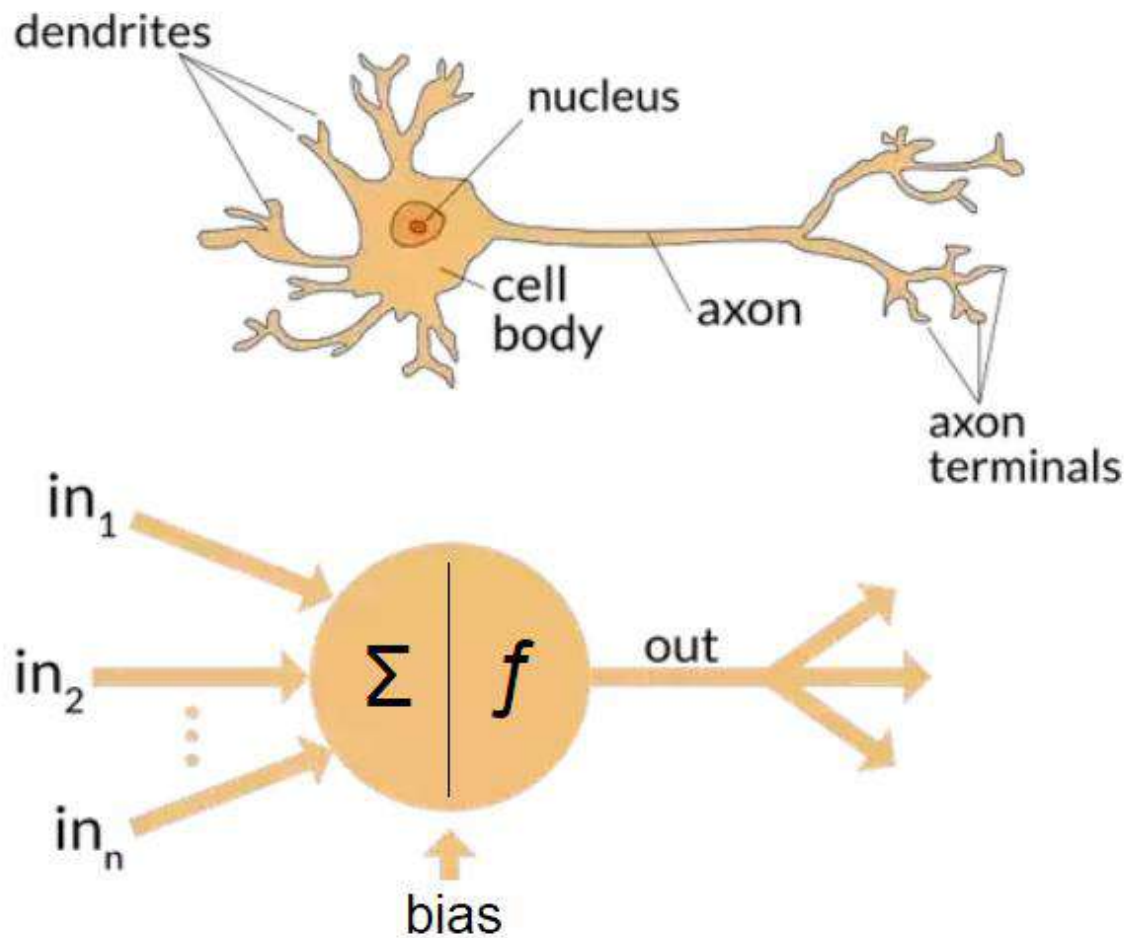
Output layer

- Output layer is used to provide outputs
- Bi classification : One neuron
- Multi classification: Multiple neurons
- 10 classes : 10 neurons
- Each neuron has bias and weights

3NN: $ip\ 1 \implies h\ 1 \implies o\ 1$

4NN: input layer $\implies h1 \implies h2 \implies$ output layer

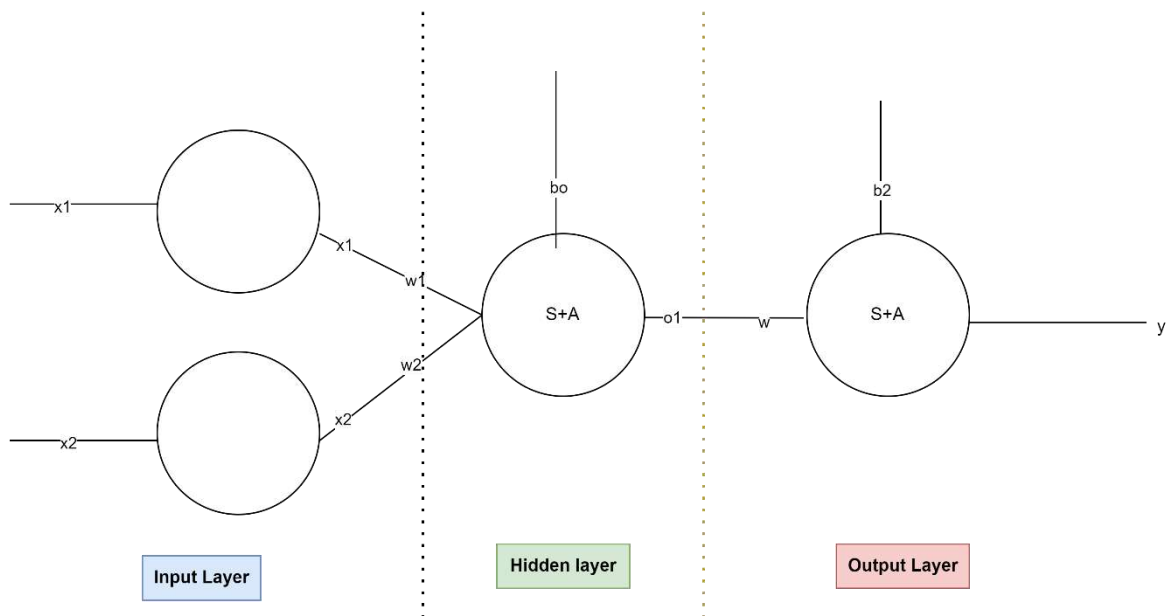
10NN: input $\implies h1 \implies h2 \dots \dots h8 \implies$ output layer



Example – 1:

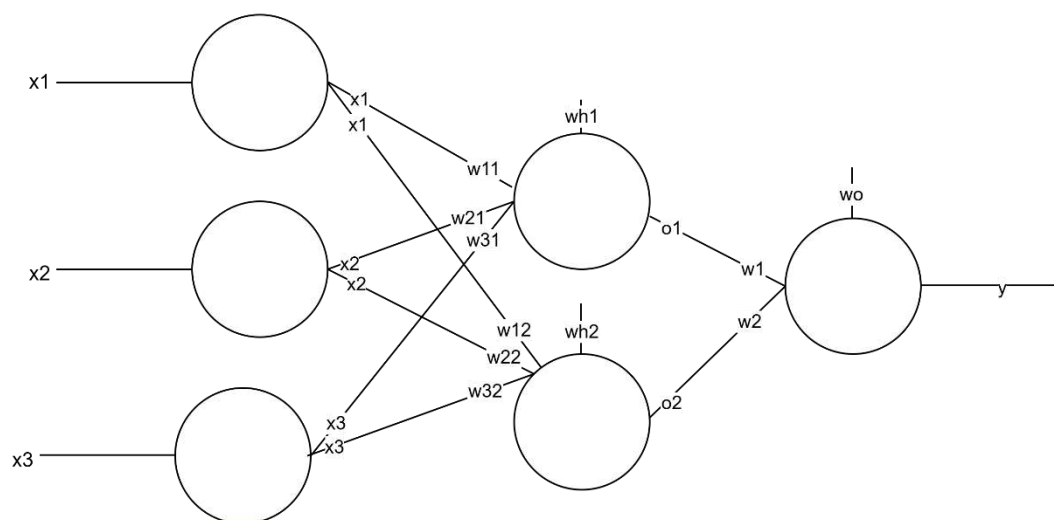
3NN:

- *input layer : take two inputs and one bias
two Neurons*
- *Hidden layer: one neuron*
- *Output: one neuron*



$$O1 = \text{Activation}(b_o + x_1 * w_1 + x_2 * w_2)$$

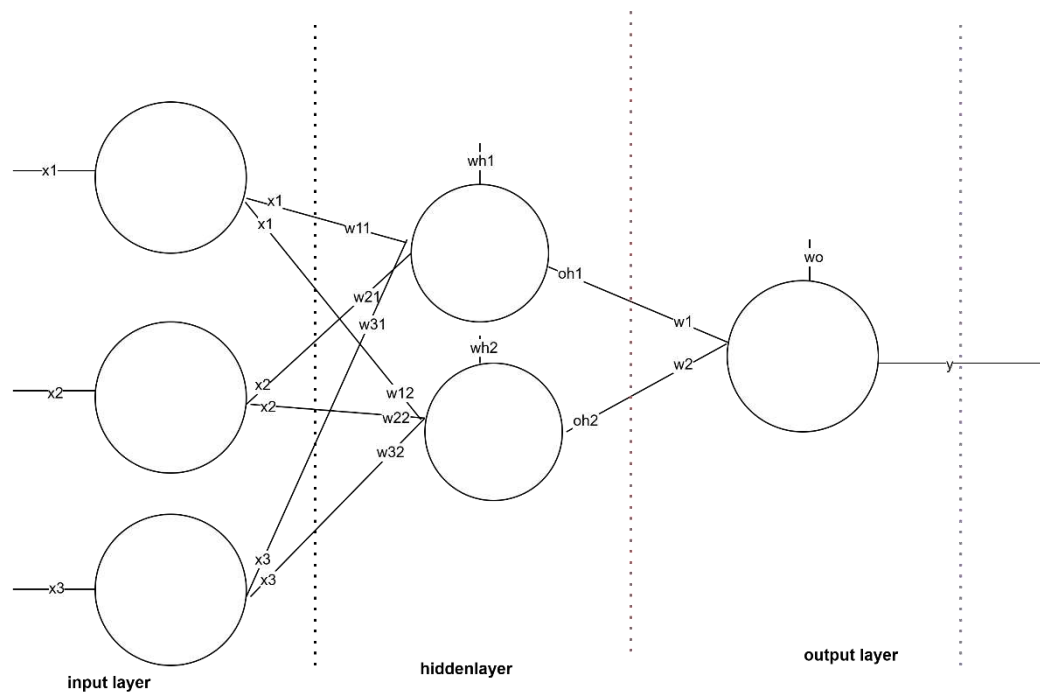
$$y = \text{Activation}(b_2 + o1 * w)$$



$$y = \text{activation}(wo + w_1 * o1 + w_2 * o2)$$

$$o1 = \text{activation}(wh_1 + x_1 * w_{11} + x_2 * w_{21} + x_3 * w_{31})$$

$$o2 = \text{activation}(wh_2 + x_1 * w_{12} + x_2 * w_{22} + x_3 * w_{32})$$



$$oh1 = \text{activation}(wh1 + w_{11} \cdot x1 + w_{21} \cdot x2 + w_{31} \cdot x3)$$

$$oh2 = \text{activation}(wh2 + w_{12} \cdot x1 + w_{22} \cdot x2 + w_{32} \cdot x3)$$

$$y = \text{activation}(wo + w_1 \cdot oh1 + w_2 \cdot oh2)$$